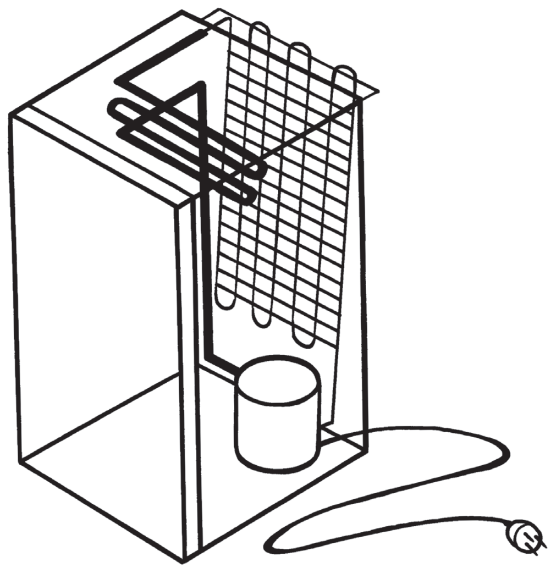
How does the fridge work? 19a

Of course you know that the fridge is cold on the inside. But did you also know that a fridge is warm at the back on the outside? How does the fridge get so cold inside? And why is it warm behind the fridge?

In most fridges, there is a system of pipes on the inside and the outside. The inside pipes are usually fused onto a silver coloured metal plate. Outside at the back, the pipes are connected to many small black metal plates. Where the pipes enter the interior, there is an expansion valve. There is also an electric motor, which is usually under the fridge at the back. This motor runs a compressor.

Inside the pipes is a refrigerant. In old fridges this is Freon, which boils at -30°C. But Freon is one of the gases that destroy the ozone layer in the atmosphere and so it is no longer used. Old fridges should have their Freon taken out professionally. Freon destroys ozone molecules in the atmosphere and so they cannot absorb the dangerous and skin-cancer causing ultraviolet rays from the Sun. These days, different refrigerants are used; for example, ammonia which boils at -32°C and HFC which has a boiling point of -27°C.

1 Use the information above to label the diagram of a fridge.



2 The ordered list of statements below describes what happens in the pipes of a fridge. Cross out the incorrect words in brackets.

a The compressor pumps the gaseous refrigerant into the *[black / silver]* outside pipes.
b Pressure in the black *[inside / outside]* pipes increases and space decreases.
c The *[gaseous / liquid]* refrigerant particles slow down.
d The kinetic energy of the particles in the black outside pipes becomes *[less / more]*.
e The 'lost' kinetic energy is changed to *[cold / heat]* energy.
f The gaseous refrigerant turns to a warm *[liquid / solid]* refrigerant.
g The black fins around the black outside pipes give off *[gas / heat]* and feel *[cool / warm]*.
h The pressurised liquid refrigerant is forced through the expansion *[pipe / valve]* to the silver inside pipes.
i The *[black / silver]* inside pipes are practically empty and have *[high / low]* pressure.
j The liquid *[air / refrigerant]* particles start to move really fast in the silver inside pipes.
k The liquid refrigerant becomes a *[gas / solid]* in the silver inside pipes.
l The needed *[kinetic / heat]* energy that is taken by the now faster moving particles comes from heat energy.
m The silver inside pipes become *[colder / warmer]* and colder.
n The *[air / heat]* of the air inside the fridge is taken by the very cold silver inside pipes.
o The fridge becomes *[colder / warmer]* inside.
p The silver *[inside / outside]* pipes become a little warmer.
q The gaseous refrigerant is pumped out of the silver inside pipes by the *[compressor / valve]*.

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3 The table below summarises what is happening inside the pipes of a fridge. Use the word bank to complete the table. Note that some words appear more than once.

Word bank

- cold
- gas
- high
- high
- high
- large
- low
- low
- low
- liquid
- small
- very fast
- very slow
- warm

Aspect of refrigerant	Immediately after compressor	In the black outside pipes	Immediately after expansion valve	In the silver inside pipes
State of matter	Gaseous refrigerant condenses into liquid refrigerant	**a**	Liquid refrigerant boils into gaseous refrigerant	**h**
Particles' speed		**b**		**i**
Pressure		**c**		**j**
Space		**d**		**k**
Kinetic energy		**e**		**l**
Heat energy		**f**		**m**
Temperature		**g**		**n**

4 As the refrigerant moves from the black outside pipes to the silver inside pipes and then back again to the black outside pipes, it undergoes changes. Write sentences describing these changes, in terms of each aspect listed below.

a State of matter:_____

b Speed of particles:_____

c Kinetic energy of particles:_____

d Heat energy of refrigerant:_____

e Pressure inside pipes:_____

f Temperature of pipes:_____

5 Complete the following sentences to explain the events in the silver inside pipes of a fridge.

a The silver inside pipes feel cold, because the refrigerant _____.

b The refrigerant boils in the silver inside pipes, because its particles _____.

c The particles gain kinetic energy, because they move _____.

d The particles lose heat energy, but they gain _____.

6 In your workbook, write similar sentences to those in question 5 for the events in the black outside pipes.

7 Finally, in your workbook write an explanation about how the inside of a fridge becomes cold.

- Write a brief **title**.
- In the short **introduction**, summarise how the fridge works.
- In a brief **description**, list the important parts of a fridge and where they are.
- In the **explanation sequence** write what is happening to the refrigerant and its particles; refer to kinetic energy and heat energy.
- Finally, in the **conclusion**, write how your life is better because of the fridge.

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